

# FUTURE

combined with **SCIENCE FICTION** stories

**JULY**

20¢

## DARK RECESS

by George O. Smith

## ULTRASONIC GOD

by L. Sprague  
de Camp

and many other  
thrilling stories

A  
DOUBLE-ACTION  
MAGAZINE

**ALL STORIES BRAND NEW**





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# **DARK RECESS**

**By George O. Smith**

**(author of "The World-Mover")**

**FEATURE NOVEL OF COSMIC SECRETS**

**Clifford Maculay was the one man who could explain the curious shift in the universe, which was far more than the academic matter it seemed to be. But Maculay had been "cured", and was no longer interested....**

**There are two basic ways to treat personality difficulties. One: change the personality. Two: remove the psychic blocks which are at the root of the trouble. The first method may be simpler, in some cases, and may be accomplished without apparent**

**harm. But what if an individual's  
worth to society is so entangled  
with his personality troubles that  
when you change the latter, the  
former disappears, too?**

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Clifford Maculay reacted instantly to the doctor's question; he became half-angry, completely indignant.

Doctor Hanson smiled. "You're not angry at the question," he said quietly; "you're not even surprised that a man of seventy should ask such a question. What you are indignant about is that your mind denies such a need. Cliff, you're trying to run your body with your brain."

"Naturally. So what has my love life—?"

"You've got glands too," remarked Hanson. "And some of them are damned important to mental balance."

Maculay sat forward on the chair, tense and alert. He was not accustomed to being browbeaten; Maculay gave the orders and other people jumped. Now that he was on the receiving end of the deal, he was preparing for the battle of wits. But Hanson had seen many such men in forty-odd years of medicine. Hanson did not see Maculay the Mind; he saw a man of thirty-eight, soft from lack of exercise, underweight from the constant burning away of nervous energy. He saw a fine physical machine being run into an early grave or a sanatorium, because the mind behind those sharp blue eyes was too damned ignorant to understand that it could not trade the worn-out body for a new model with white sidewall tires, automatic defroster, and long-playing record attachment.

"Relax," said Hanson; "I'm not going to argue with you."

"Good. Now let's get down to business."

"Exactly what do you want?"

Maculay pondered for a moment. "Do you understand variable-matrix radiation mechanics?"

"Probably as little as you know synaptic pressure theory."

"That's the trouble. I can't explain in detail what I want. I can only explain by analogy. Look, Doc, for eight years I've been experimenting with some mathematics along an entirely new field of theory. Indications are that gross matter can exceed the velocity of light under certain conditions; but in attempting to define these conditions by mathematical formulation I've hit a snag."

"What manner of snag?"

Cliff leaned back in his chair and stared at the ceiling. He was physically relaxed, now, but only Doctor Hanson could hazard a guess as to how much of this man's metabolism went into the job of keeping that big brain in high gear.

"Physical matter cannot, of course, exceed the speed of light in universal space. However, normal space is no longer normal when it is warped by electrostatic fields, electromagnetic flux, or gravitational lines. These universal effects produce a warping of physical space to such an extent that the warped area is no longer a part of, or connected in any way with the universal space we know. It becomes a small island of separate space which may be accelerated or retarded. That's the snag, Doctor."

"I don't see it."

"I always end up with one equation that has two answers. Theoretically, one must be real and one must be imaginary, somewhat like the solution to a simple quadratic; in that case you can disregard the answer that tells you that you are confronted with a minus quantity of mass, for instance, and you can select the positive quantity as being correct with neither difficulty nor ambiguity. In this case, being more complex by far, I find two roots indicating a positive and negative space, mutually inimical. And, what causes the trouble is the fact that the determinant depends upon the development of a negative-gravitic field."

"Well—?"

Maculay laughed bitterly. "This is sheer nonsense; like dividing by zero."



Hanson shrugged. "So?"

"Obviously I have made an error."

---

Doctor Hanson again shrugged, wondering what the man was getting at. Electrical engineers confronted with a tough problem in vector analysis consulted other electrical engineers; they did not bring their unruly vectors to a psychiatrist or physician and hope to have them solved. They came to medicine and psychiatry when they began trying to integrate and plot the rhythmic sway of their secretary's hips, or began to see the outline of a woman's lips in the catenary of a suspension bridge.

"Obviously," nodded Hanson.

"So here it is again, Doctor. I've been back and forth across my equations for the past eleven months and always come back to the same errata."

"But what can I do?"

"Someone must check my equations—someone who is viewing them as a competent, but unbiased, observer."

"An excellent idea."

Maculay spread helpless hands wide. "I sound like an egomaniac," he said, "but there is no other man on earth who can follow my mathematics but I."

"Not even the thirteen fellows who understand Einstein?"

Maculay snorted. "Understood," he said emphasizing the last syllable. "Einstein was difficult when first made public; nowadays there are plenty of men who know more about Einstein's theories than the man himself. In my own case it is similar. No other man has had a chance to study my theories; I have a few adherents who try to follow them, but they have not the full time to put to the job and so they are far behind. Besides, I'd trust none of them."

"I see."

"Ergo, Doc, what I say is this: You are to hypnotize me. You are to give me a post-hypnotic suggestion that I am to forget the error in my calculations,

that I am to recheck them carefully and completely, without knowing that some factor in them is in error. Then and only then can I locate it; as soon as I locate this error, I am to remember everything."

"Supposing your mathematics is not in error but is entirely correct—suppose no error truly exists?"

"There is that possibility; but if the paradox is true, I will have at least been forced to forget that I once believed an error existed. But I must check this math as a competent and unbiased observer."

"That can be done."

"Good; now let's get going and let's have no more nonsense about my glands."

"This I can do; I will help you."

Maculay relaxed while the doctor produced his hypnoscope and set it up on the table. With Maculay's cooperation, he was in the hypnotic trance in a matter of seconds.

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Doctor Hanson looked at the man. This was probably the first time that the entire man had relaxed, mind and body, in years. But Hanson did not see the point: Maculay may have run into a mathematical paradox, but it was not of honest mathematics. Figures do not lie, but liars can figure; it is more than possible that a brain will introduce an error in order that the facts of the case be unrecognized. Hanson nodded quietly. Man was mind and glands and body and appetite, bones, hide, and ulcers. If a sick mind can produce a sickness of the body, the reverse is true. Cliff's error was not in his mathematics; it was in his life.

Of all the things Maculay needed, more work along the same line with no relaxation was not among the list. What Maculay needed—or would eventually get in a sanatorium—was a long period of relaxation. Fun and games; a bit of competition; a hangover, and the sheer physical delight of wrapping an arm about the slender waist of a female and swaying to and fro to the rhythmic beat of tomtoms and the howl of a well beaten clarinet.

At seventy, Jay Hanson had learned the impatience of youth. Maculay had a lot of time to finish his equations. Scissor a year of Maculay's life and he could then finish this; let him go on as he was and he would burn himself out at a mere fifty.

He looked at Maculay seriously. "You have been working too hard," he said.

The reply came instantly, like the echo of an automaton.

Hanson nodded to himself. It was obvious; when the burning drive of that demanding brain was stilled, the subconscious recognized the fact that Maculay was working too hard.

"Maculay's Equations are in error," said the doctor.

Cliff Maculay stirred, shook his head, and began to disagree violently. Then he relaxed, since he had come there to solve an error; but he had become tense again.

Hanson shook his head unhappily; this was going to take time and effort. He must take this conversion slowly, since it was apparent that the slightest touch upon dangerous ground would trigger the big brain into reaction and perhaps undo in the space of a second the work of several hours.

Gradually, prying and working, Hanson began to elicit information from Maculay. Bits of character traits, an impulse suppressed, an attitude formed in youth, an impediment created to shut out the demands of normal living, desires for this and wants for that. Hanson looked at them clinically, then either reversed them or let them stand, depending upon their possible affect. Each phase took time; it is not simple to take a man who has never held a billiard cue and make him believe that excellence at the pool table is an evidence of a sharp eye and fine coordination instead of the result of a misspent youth. And Cliff's attitude towards women was troublesome. His mother, the youthful reading of too much of King Arthur, or Lord knew what, had given Maculay the odd idea that a woman was a sort of goddess, not to be touched by the hand of clod-like man. To reverse this attitude towards a more practical attitude was difficult, since the reversion must not be complete. Hanson did not want Cliff to reverse completely, to the other extreme, where the man would go out and start treating women like galley

slaves, punching bags, or chattels—which, in fact, was about the way Maculay had expected to be treated.

Hanson took a brief rest from the hard job, by recalling and telling Maculay every risque story he could remember.

Then he was at it again, prying and probing, and reversing Maculay's attitude on gambling, liquor, tobacco, and politics. He made a slight revision on Cliff's idea of proper dress; the physicist had a horror of appearing dirty, even when engaged in the dirtiest of jobs. With some effort the doctor convinced Maculay that a mechanic emerging from beneath a car with a face full of grime was not automatically an undesirable character, either to men or women. The crux of the matter was whether he liked that condition of dirt or not.

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With a number of factors accomplished, Hanson took a deep breath, felt his pulse, counted his heartbeat and respiration, and fished for a pill from his desk and swallowed it quickly before he went on. The hardest part was to come.

Cliff took himself seriously, far too seriously. With delicate verbal barbs, Hanson began to poke fun at some of the imbecilities of pedantic reasoning. Maculay offered resistance at first, but Hanson worked him over the ground carefully, pointing out that Maculay, the only man in the world capable of understanding the variable-matrix wave mechanics, was in no position to snort at his fellow man. After all, Gertrude Stein had once gained great popularity on the theory that no one could understand her and therefore she must be sheer genius. Eventually he had Cliff laughing over an old limerick:

*The wonderful family Stein,  
There's Gert, and there's Ep, and there's Ein.  
Gert's poems are bunk;  
Ep's statues are junk,  
And nobody understands Ein.*

Hanson worked over Maculay's Equations with a bit of acid humor. In third person, he had Maculay chuckling over the physicist who worked for years on some mathematics that did not come out even. Gradually, the doctor convinced his patient that he was not Clifford Maculay, the renowned abstract mathematician, but Maculay's nephew—the black sheep of the family—who viewed the brainy members with as much distaste as they viewed him. Young Cliff had often been mistaken for his brilliant uncle, and found this funny, since he felt himself smarter than his namesake; he, young Cliff, had fun whereas his uncle had only hard work to show for his life. Actually, any pondering of his uncle's work made young Cliff sick to his stomach, and he was glad to ignore such things; the whole theory was so much stupidity.

And for one year, Clifford Maculay, physicist, would be as different from his former self as was possible without breaking the law to bits.

"At the end of this year, you will return to your apartment in Washington, take a good night's sleep, and awaken as Doctor Clifford Maculay. Then, and only then will you remember; and you will realize furthermore that this job of relaxation has been forced upon you for your own good. You will then be able to solve the error in your calculations."

Hanson paused for a moment, pondering as to the advisability of giving the hypnotized physicist a key-word to bring him out of the post-hypnotic suggestion. But Doctor Hanson was seventy years old; he knew all too well that a year from this moment he might be dead and gone. He viewed it calmly, but not disinterestedly, and decided against a key-word; it only introduced a conflicting factor.

Let the man awaken of his own accord.

Then he awakened Maculay, who sat back in his chair with a chuckle, reached for a cigarette from the box on Hanson's desk, and puffed at it with relish.

"How do you feel?" asked the doctor.

"Like a million," said Maculay.

"Good. Come back in one year. I'll have my girl make an appointment. For now, we're all finished."



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Doctor Hanson stood and watched Maculay head for the door; the physicist's step had a certain bounce, curtailed by the fact that the unused muscles of his body were not used to the catlike stride of the completely balanced, healthy man. A few days of that sort of bounce and Maculay would have it. The door closed exuberantly and Cliff was on his way to a one-year binge.

He paused once outside of the doctor's office. Ava Longacre was bent over some notes, and Cliff viewed her contemplatively. She stood up and smiled at him. It was a sort of professional smile, the kind she gave all of the doctor's visitors; it made no difference to Ava whether the visitor were seventy or seventeen. She gave each of them the same dry smile.

Cliff crossed the office in a quick stride and put both hands on her shoulders. He drew her forward, felt her instant stiffening relax; with a cheerful upsurge of spirit he put an arm around her, tilted her face upward with his free hand and kissed her. He felt her yield to him, press against him softly, then respond.

Cliff knew he could have her, but in that moment he also knew that he really did not want her. Ava was a bit over thirty; she had a quiet, mature quality—good-looking, but far past the radiant flush of youth. A hard-working woman, efficient, intelligent, Hanson's nurse, medical aide, and receptionist, did not offer the fun and frivolity that Cliff Maculay sought.

He stepped back and smiled down at her. "Nice," he said with a chuckle. Then he kissed her again, lightly on the mouth, turned, and left the office.

Her cheeks burning, Ava Longacre stamped into Hanson's office.

"What goes on?" she demanded. "What on earth did you do to that man?"

"Why?"

"He came in here like the proverbial absent-minded professor, his eyes blank and sort of muttering to himself about radiation mechanics or the like. He didn't even look at me."

"Then?"

"On his way out he sort of grabbed me and kissed me."

Hanson nodded appreciatively. "You liked it?" he chuckled.

Ava sat down, landing in the chair with a thud. "When a man puts a hand on me and my knees turn to jelly," she said quietly, "I oscillate madly between hating his guts and wanting him to try it again. That sort of thing would play hell with a girl's morals."

"Shucks," chuckled Hanson. "I've just violated all of the rules of medicine. I've just treated a man against his will—and turned an introvert inside out."

"You sure did," nodded Ava.

"He'll be back again in a year—and normal, then."

"But how do you turn an introvert inside out?"

"Reverse his sense of values."

"But—"

"His memory pattern? That's difficult. To make him more or less stable for that year, I sort of tampered with his memory on a temporary basis, also. He thinks he is all sorts of things that he has never been—but has probably wanted to be from time to time."

"Is that why he kissed me?"

"Partly. But you're the woman he should have when normal, not as he is now. That's—"

"So you gave him a false memory, complete with a lot of details to explain just about every possible question, hey?"

"Yep."

"And just how was this background furnished?" she demanded.

"Remember it is only temporary and need not be complete. Just sufficient to justify its being."

"Don't quibble."

Hanson laughed. "Well, when a man of seventy starts to furnish a bit of background for a youth of thirty-odd, what better than a few true experiences from the old man's past."

Ava Longacre snorted. "I'll bet you were a hellion in your youth," she snapped. "And in your old age you're a nasty old lecher."

Hanson squinted at her. "I wish I were forty again," he leered. "But worry not, m'lady. Maybe the basic idea was mine, but Maculay kissed you on his own account. And I commend his taste."

Ava uttered a single, explosive "Oh!" and stalked out angrily, slamming the door behind her. She leaned against the hardwood panels and listened to the roar of Hanson's laughter die in a slow gurgle. She pegged it properly as part hysteria; the hours of hard mental effort spent on Maculay would have taken a lot of pep out of the Old Boy, and he would then clutch at anything remotely amusing and make an uproariously funny incident out of it. But this was not funny.

She remembered Maculay's hand on her, and her body went supple against the door. Then by sheer mental effort she snapped her head erect and walked from the door, determined to forget it.

Ava did not recognize the fact that for hours, days, or months—and perhaps forever—she might be telling herself that it was a good thing to forget about.



The *Island Princess* took off on schedule, arrowed into the blackness of space, and set her nose-sight on Venus. She was forty hours a-space when it happened. And the *Island Princess* was one of the four spacecraft close enough to the thing to have its presence recorded in the celestial globe.

It came with a roar of sound from the radio, which eliminated all communications instantly, and continued on a diminishing power for an

hour until it fell below the cosmic noise level. It appeared in the celestial globe as an ebon shaft; measurements made it a half mile in diameter but extending from beyond the range of the globe in both directions. It was as straight as it could be. On the other ships, the same facts were noted.

Upon the several planets of the solar system, cosmic-ray counters went crazy. Showers of unprecedented violence bathed the solar system in a raging torrent of high-energy particles. The showers continued, diminishing in intensity as time went on; the slower particles arriving last, of course.

But the one thing that caused consternation throughout the entire solar system was a sickening *shift*. Spacecraft and planet gave a tiny, queasy slip, sort of like a heavy man who has just trod upon a grape. Things move according to their mass and according to their distance from the black shaft of energy. On planets, it was just enough to cause fear; the most infinitesimal waver in the constant course of the planets awakens racial fear. In spacecraft, the shift was more violent, but here people were prepared for a bit of wobble.

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**On planets, the shift was just enough to cause fear; auroras flamed for an hour....**

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The auroras flamed bright for an hour; as soon as the shift had shaken the wits out of every human in the solar system, the big observatories set their big glasses on the fixed stars and consulted quadrant-protractors to ascertain what the shift had done. On photographic plates of operating telescopes, the shift was barely noticeable upon the images of brighter stars. The dimmer ones had danced aside and back too swiftly for the emulsion to register. But it had been a swift jiggle up and back; now things were as they had been once more save for the big mystery that caused the radio lanes to buzz, and made men ask their neighbors what it could have been.

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Captain Bardell of the *Island Princess* went to the salon after the radio had told him what little could be told about the incident. He told the passengers what he knew as a matter of interest, and because they had been as close to the phenomenon as any other human being.

Cliff Maculay, bent back across the bar with his elbows hooked over the edge and a glass in his right hand, chuckled amusedly.

"But this isn't funny," complained a comely woman in a strapless gown at his left.

"Dorothy, darling, you're wrong," he laughed.

"I suppose you know what it is?" came the cynical reply from the red-head on Cliff's right.

"Helen, that 'thing' was a manifestation of the application of variable-matrix wave mechanics to intrinsic space by the real and/or unreal roots of negative space."

"That's utter gibberish."

Maculay laughed. "Verily," he chuckled. "But my revered Uncle Clifford will—about now—be telling the world the same thing in about the same incomprehensible collection of dictionary fodder."

Captain Bardell heard, and came to stand before Maculay. "Clifford?" he said uncertainly. "Clifford Maculay?"

"Right name but the wrong character," said Maculay, sipping from his glass. "Doctor Clifford is the genius in the family; Cliff, the nephew, has only genius for getting into mischief without getting into trouble about it. To each his own," he chanted, lifting his glass in a toast.

Bardell was openly disappointed. "I'd hoped you might give us an idea of what was going on."

Maculay turned, rapped the bar with the heel of his glass to get attention, and then turned back to the captain. "I can," he said cheerfully. "But do you have the faintest idea of why nephew was relegated to the Outer Darkness?"

Everybody, listening to Cliff, shook their collective heads.



Maculay laughed. "They had me studying under him for years. Doctor Maculay is a slave driver and a martinet. Cigarettes, liquor, and wild women are annoying things that detract from the single-purposedness of life. Doctor Maculay is the kind of duck who would rather work overtime than make frolic with a dame—and he expects everybody who works with him to do the same. He also pays them accordingly, since a small room, a sterile diet, and a minimum of clothing are all that is necessary for any man dedicated to science."

"So?" asked Bardell, a bit angry at this man for belittling one of the solar system's greatest minds.

"So Cliff, the ne'er-do-well, used to take a few of Uncle Clifford's well-flanged ideas, add a character, stir well with a villain and a dame, and emerge regularly with a bit of science fiction. I was Ed Lomax, one of Larimore's cover names until John used the right name instead of the pseudonym, and people started to write fan letters to Clifford Maculay, MM, PhD, et al. Shortly afterwards I was out of a job."

"Then you do understand some of this?"

Maculay grinned and nodded.

"But Doctor Maculay will be able to figure this thing out?"

Cliff nodded again, and smiled. "Good thing, too," he chuckled. "He is the only man in the system that can handle it without going off half-cocked. Maculay may be a stuffed shirt but he is no imbecile. Tinkering with inverted space—or pouring a quart of nitric acid over a half-gallon of glycerine—might be deadly unless you understand what you're doing. Maculay is super-cautious about anything that he does not understand completely. I cannot say the same for his underlings, who casually point out that mankind had been using electricity for years and years before they knew anything about it. But," he said with a laugh, "enough of the manifestation of the unreal roots of variable-matrix wave mechanics. Maculay's wastrel—but interesting—nephew is about to enjoy life."

Cliff winked at Dorothy, patted Helen on a bare shoulder, and then led Alice towards the dance floor.

"Doesn't this mean anything to you?" she asked him.

"Uh-huh," he said with a smile. "At about three cents per word; that black shaft of energy is an idea coming to life."

"What kind of idea?"

"Um—let's see. That black shaft of energy was really a spacecraft, passing through the solar system at a velocity higher than the speed of light. Some extra-solar race, colonizing the galaxy. What we detected was the space-wake of such a craft. You have no idea of the energy kicked up when a body passes through space at a velocity higher than that of light. Then Our Hero, bullied by his superiors, shows that he has measured the energy-curve and solved the secret of interstellar travel."

A slight frown came to Maculay's face. "The trouble is that this super-galactic race has learned how to create negative space before the ship and re-create positive space behind it to keep from having the—the—" A bead of sweat came upon Maculay's face and he became nervous. He looked around, almost wildly, before continuing, "the—entire universe," he concluded lamely. "Negative space is self-propagating, you know." Maculay finished this last with a wince of pain.

Then Maculay straightened up with a laugh. "That's lousy," he said. "Larimore wouldn't buy it. We'll have him go out and meet some four-armed monsters who think that human meat is superb. That's crummy too, but it's an idea. C'mon, m'lady, let's dance!"

---

The telephone rang on Doctor Hanson's desk. It was Ava, from the outer office. "Man by the name of Redmond to see you, doctor," she reported.

"Has he an appointment?"

"No, that's why I'm calling. He claims it is a matter of impor—No, Mister Redmond, you cannot go—"

The doctor's door opened abruptly and the man called Redmond strode in. "Where is Maculay?" he demanded sharply.

Doctor Hanson looked up at Redmond calmly. With insulting deliberation, Hanson eyed the man, while Redmond began to fume. Redmond was tall

and thin, a bit too tall and a bit too thin in the doctor's estimation. He was thirty to Maculay's thirty-eight, but did not smoothe his impatience and ambition behind a cloak of politeness.

"Sit down, Mister Redmond. I'm interested in you."

"Where is Maculay?" came the repeated demand.

Hanson smiled slowly. "I'm interested in trying to discover just what it is about abstract mathematicians that makes them think that they can stamp their way through life, disregarding not only the rights of others, but their own as well."

"Enough of this damned foolishness—"

"Shut up, you young whippersnapper!" roared the doctor in a voice that rattled the windows. Redmond shut up. "I'll have respect from you, Redmond. And if I don't get it, you'll leave. Understand?"

Redmond bristled.

"Relax," said Hanson. "I'm no longer able to punch your face as you request by your actions, but I know several men who would be most happy to help me in this matter. Now, what is it that you want?"

"I want to know where Maculay is."

"I don't know."

"Damn it, you do know."

"Redmond, I'm not a liar."

Redmond leaned forward over the doctor's desk. "Maculay came here," he said, "and I know why. Maculay did not return from here, and I want to know why."

Hanson leaned back in his chair. "Doctor Maculay came here and discussed his difficulties with me," he said. "During the course of the discussion, it became quite evident that Maculay was on the verge of a nervous breakdown because of too much hard work and too little relaxation. I convinced him that a long vacation would enable him to live and be productive longer than he might enjoy if he went back and killed himself on his job."

"So where did he go on this vacation?"

"Maculay admitted that if a single soul knew his address, they'd be sending him problems within a week. He took off, destination unknown."

"Did he say when he would be back?"

"In one year."

"A year! My God! We can't wait!"

"Can't is an impossible word," remarked Hanson.

"But we must find him."

"You might start combing the solar system," suggested the doctor.

"Impossible. Yet—"

"Redmond, there have been many indispensable men in history who were not so indispensable that their leaving caused affairs to stop short. I admit that their plans often flopped, or that history took a little longer to get itself made when their driving force died. But not a man on record has ever been truly indispensable."

"But you don't understand," complained Redmond. "It's about that blast of energy that shocked space."

"I guessed as much. What is—or was it?"

"We don't know; Maculay does, or can deduce its meaning."

---

Redmond started to stride up and down the office at this point, talking half to himself and half to the world in general. Through the still open door. Ava could hear him, and since the danger of attack had been averted, she decided to close the door. But Hanson waved her inside where she sat in one of the inconspicuous chairs along the far wall. Both Ava and the doctor watched Redmond quietly.

"From what little we know of it by direct observation it came all at once—a shaft of energy as instantaneous as birth. Where once was empty space, this bolt of energy was created. The energies it created showed no directional

qualities, and it extends as far in either direction from here as we care to imagine. The distant energies are still coming in from both directions, diminishing because of the tremendous distances, but still showing nothing directional."

"But this shaft of energy must have come from somewhere?"

"Did it?" exploded Redmond. "Did it come from somewhere—or did it burst into being instantly from one end of the universe to the other like the creation of a rope from nothing all along its length? Actually, we know this: Its duration was as close to instantaneous as anything might be. The rest of the phenomenon was merely persistence of the energies it created."

"Those are questions that I cannot answer."

"Maculay could."

"You assume that this thing might be in Maculay's field?"

Redmond nodded. "We charted the energy-curve," he said. "Then one of the boys integrated the curve and came up with a formula for the curve which I saw and without any trouble at all reduced to one of Maculay's Equations. Do you know what this means?"

"No. Of course not."

"This means that the validity of Maculay's Equations is proven fact. Just as Maxwell's Equations were proved by the existence of electromagnetic waves in nature, so are Maculay's Equations proved by the existence of this manifestation of the real and unreal roots of space, occurring in nature." Redmond resumed pacing again. "What is maddening," he said, "is the fact that we do not know where it came from."

Hanson shrugged. "You said it sort of leaped into being."

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Redmond paused and beat one fist down on the doctor's desk for emphasis. "So it seems to our blind, deaf, stupid senses who plod along the universe limited to the speed of light or sound. Man—the fleet bullet snaps past your

ear at a speed faster than sound. Can your limited senses tell me whence it came?"

"Yes."

Redmond shook his head. "Not from the sound of the *Snap*!" he snapped. "You tell from the sound of the gun—which comes later! With a silencer, you would be unable to line the flight up. So," said Redmond, staring at the wall again, "something fired a bolt of energy that propagated faster than light, creating its own negative space as it passed. What it was we shall never know. But—" and he bored at Hanson with sharp eyes,— "get me Maculay and we shall follow it into interstellar space!"

"And if Maculay were to die tomorrow?"

"Then we would follow it sooner or later anyway. A bit more fumbling, a bit more walking an unfamiliar way in the dark, but we would get there." Redmond looked at the doctor solemnly for a moment. "A year?"

"A year."

"Hell," snapped Redmond, "in a year we can do it ourselves! A year! Hanson, Cliff Maculay has always kept a volume of data from me. I am going to open his desk and get that volume, and go to work on this thing myself. Were he here he would forbid me, but he is not here. I—"

"Why are you telling me this?" asked Hanson.

"Someone must be told, and I—" Redmond trailed off uncertainly. Then he nodded and left the office as abruptly as he had come in.

Ava blinked. "What do you make of him?" she asked uncertainly.

"Very simple. His is the case of not-quite-genius working at the feet of true genius. His pattern is poorly aped after Maculay's forcefulness, but obviously lacks Maculay's weight. He wants to give the impression that he is cut of the same cloth as Maculay. He is uncertain of himself, or he would not bluster and threaten; nor would he be so completely at sea without Maculay. He has a frustration; Maculay's secret data has been withheld from him. He is jealous of Maculay and also fears Maculay or he would not make a confession to me that he was about to break orders. Furthermore, he is convinced that he can solve this thing without Maculay's help, but wants other people to believe it also."

"But could he get into trouble?"

Hanson laughed tolerantly. "Any man who has lived beyond the age of eighteen months can get into trouble," he said. "And it's good for a man to get into a bit of trouble occasionally. Security is a fine goal, but it is danger that sharpens the wits and eventually sets the character into such self-confidence that his security comes from within rather than without."

He leaned back in his chair and thought for a moment. He was not quite correct in telling Redmond that he understood nothing of Maculay's work. During the hours Hanson had hypnotic rapport with Cliff, he had absorbed quite a bit of Maculay's theories. Not that Hanson could stand in Maculay's shoes—or even his baby slippers for that matter—but he had a fair idea of what Maculay was driving at. He took this on faith rather than a real understanding—as any man might nod his head and accept the formulation of the three degrees of infinity because some bright man told him that such existed, one still might not understand why the number of spots on a line and the number of spots on a plane—when a plane has an infinite number of lines and each line an infinite number of spots—were both of the same degree of infinity.

Something niggled at Hanson's mind—something important in just plain horse-logic that had come to him fouled up in a barrage of words and formulations that were so much triple-talk to the man untutored in abstract theory on variable-matrix wave mechanics. In the maze of completely confusing theories, it was like the sighting of a shaped stone arrowhead in the rubble of a landslide, or finding an empty tomato can lying on the absolutely barren and completely useless fifth satellite of Saturn.

Someone had shouted "Two plus two equals four" amid the babble of an insane asylum, and it made sense.

Hanson ordered his whirling thoughts, marshalled them as only a man who has pursued the mysteries of the human mind for fifty years could do, and made his recollections come out consecutively.

And then he hit the desk with his hand. "Negative space depends upon the generation of a negative-gravitic field," he muttered. "Which produces the unreal root, and positive and negative space are mutually inimical."

"What was all that?" asked Ava.

Hanson shook his head. "Damn it, Redmond is right. We *need* Maculay!"

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Ava stared at the doctor. "But...."

"Ava, from what I gather, Redmond is about to get into the production of a negative-gravitic field, which will generate negative space, which will destroy this space. Doubtless that shaft of energy, so called, was nothing but a shaft of negative space that met with and cancelled real space with the resulting outburst of energy."

"But—"

"We need Maculay," said Hanson solemnly. "No one will believe me, for I obviously know far too little of the facts. I admit that I am just guessing, but I have the feeling that the error in Cliff's Equations was no error at all. What drove Maculay into a mental whirly-gig was the fact that he had discovered at the end of his fingertips the ability to destroy the solar system—or destroy something equally as big. His was the shock of the child who has been playing with matches in a powder-house and discovers long afterwards just what fate he had escaped by sheer luck."

"But what are we going to do?"

Hanson smiled confidently. "We're going to get Maculay back here long enough to tell us the truth."

"But you don't know where he's gone."

"Since Cliff now has all of the instincts of a tomcat," chuckled Hanson, "all we need do is to imagine where a tomcat would go—and go there. Ava, if you were a brazen hussy, where would you go to huss?"

Ava froze. "I'm not!" she snapped, "and I wouldn't know."

"Maculay went to Venus," said the doctor, "where reformers, theologians, and politicians have not taken all of the fun, chance, and sting out of life."

"But how are you going to get him back?"

Hanson shook his head. "I'm not," he said; "no spaceline would take me. I'm seventy, a little creaky in the arthritis, a bit leaky in the pump, and a trifle



sclerotic in the arterios. I admit that I am the healthiest doddering old man on earth—but it is on earth that I shall stay."

"Then—" said Ava uncertainly. Her eyes began to widen with growing understanding and she backed away slightly.

Hanson nodded. "You're going to go get him."

"I'm not."

Hanson shook his head. "You'll be safe," he said. "At the present moment you have too many inhibitions to rouse a stir in Cliff Maculay."

Ava snorted angrily. She was still forgetting Maculay; in fact she forgot him four or five times each day. Each time she reminded herself that it was a good thing that she did not 'go' for his type of man since the two of them would never get along.

Defensively, Ava said, "I'm to go to Venus and comb the entire planet for a man on a binge?"

Doctor Hanson chuckled. "For he who knows the answer, Cliff Maculay would leave a trail a mile wide," he said. "But you'd never make the grade, Ava."

"You're quite right," she said.

Hanson grunted unintelligibly. It sounded like agreement to Ava, but was actually a grunt of disgust. The doctor was old enough to be beyond the sparring age, and he was disgusted at the sidelong mental attitude of a race that admitted that love, marriage, and a family were at the bottom of all effort—and then invented croquet, television, and chaperones to make it difficult.

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Hanson looked at his nurse, and shook his head slowly. He was willing to bet his hat that Ava remembered every line in Maculay's face. And that her dislike of Maculay was as genuine as a seven dollar bill, Hanson would also bet money on. He had not been untying mental knots for fifty years without being able to listen to one statement and hear the truth unspoken between

the words. He watched her stand there uncomfortably, and knew that she was uncomfortable because she knew that he knew what she was trying to hide to herself. Deliberately letting her squirm, Hanson began to fiddle with his watch chain.

He was thinking with the back part of his brain, now. He needed Maculay; he had here before him a girl that could, if she were willing to admit it, go forth and get Cliff. But not the way she was, with her defensive armor all set up to fight against the Clifford Maculay that had kissed her and then patted her on the head and left to go in search of beer and beauty.

Hanson fiddled with his watch chain, then began to swing the Phi Beta Kappa key around his forefinger, winding it up and then reversing it to unwind and rewind in the opposite direction.

Ava stood there uncertainly, watching him whirl the chain. She could not leave without some explanation regarding her reticence about going to Venus for Maculay. Obviously Hanson was not finished with this conversation, yet he sat there deep in thought. Ava anticipated that he would come out of it with a more practical idea than sending her for Maculay, since that would not work.

"Relax," said Hanson quietly, after some time. "I wouldn't send you after Cliff Maculay; I wouldn't hurt you for the world, Ava."

"I know," she said. "I—"

"You've been dwelling on that subject too long," he told her.

She nodded.

"Probably losing sleep, too."

"I wouldn't say that."

"But you look as though you needed sleep."

"I don't really."

"Then why are you yawning?"

"Am I?"

"You yawn frequently. You should get more sleep. Why not rest? Sit down and relax. Sleep is the great restorer; you should take a short nap. Sleep,

Ava. Go to sleep. I'll see that no one harms you. Sleep."

Her eyes fixed on the whirling watch charm, Ava slowly let herself down into the doctor's consultation chair and leaned her head against the back.

He passed a hand before her open eyes and she did not blink. With a quiet chuckle, Hanson dropped the watch chain into his vest pocket and sat back. "I'm going to help you."

"I know," she replied.

"You resent Maculay."

"I do."

"But it is true that your resentment of Maculay is because he is attractive to you—but wants a more vivacious and interesting type of woman."

"Yes."

"You also resent the fact that this desire of his is false, that any alliance he may make will also be false while a true love awaits in you, unwanted so long as he is under my post-hypnotic suggestion."

"Yes."

"Then since you and he are quite alike in so many ways when normal, if you are reversed in personality as he is, you will then match his mood and desire."

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With Ava, Hanson had much less difficulty; he had known the woman for ten years, known her moods, her likes and dislikes, and her personality. He had, lightly, worked her over from time to time until his control over her was quite complete. It took him about two hours to turn Ava's personality inside out and to suggest that she remain extroverted until Maculay was returned to earth. For travelling expenses he filled her wallet and gave her hypnotic reason for possessing that sum of money. Then he snapped her awake and watched her leave the office with a cheerful stride.

"A hell of a note when the fate of the universe depends on libido and post-hypnosis," he grunted.

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Seven hours later the *Evening Star* took off for Venus, and even Doctor Hanson might have had trouble in recognizing his nurse. Gone were the glasses, the mousy clothing, and the flat heels. From pedicure to hair-do and from hide to handbag. Ava Longacre was as changed as her personality.

And where Maculay had leaned against the bar, regaling a couple of women with idle chatter, Ava sat and watched four dazzled males vie against one another for the privilege of a dance, a smile, the purchase of a corsage or a drink—or the spacecraft itself.

She enjoyed it, but she remained a bit aloof; she had a job on her hands. She knew where she was going, and exactly how to find Cliff Maculay.



Alone in his office at night, while the *Evening Star* was starting the hike to Venus, Doctor Hanson sat thinking. He was piecing it together; and it was like playing with a jigsaw puzzle that had three-quarters of the pieces missing. He never would get the completed picture; it just took too many years of a man's life in study and application to finish the job. All he could do was to fit the meager pieces in where he thought they might fit, and then try to ignore the blank spaces that he could not possibly reconstruct.

At midnight, Hanson took to the telephone and called California.

He heard the operator say: "Chicago is calling Doctor Rober."

The switchboard girl at the far end asked: "Who is calling, please?"

"Doctor Jay Hanson."

"Doctor Rober is busy at the moment; may I have him call you back?"

Hanson roared: "I know he's busy. Tell him it's Jay Hanson and see what happens."

A moment later there came a grumpy voice: "Hullo. What's so infernal important?"

"Steve? This is Jay."

"That's what the gal said; it better be important."

"To hell with your precious telescope, Steve; I want some information."

"You'd think we had nothing to do but cast horoscopes," growled the astronomer. "Or answer damned fool questions about the end of the world."

"Answer me one more."

"The world has been here for two times ten to the ninth years at least; you'll not live to see the end of it."

"Look, Steve, this may be important. Tell me, have any of your instruments shown any difference in setting since that streak of energy went through the solar system a few days ago?"

"Not that we can measure."

"But—"

"Jay, the best information we can collect is that the original streak was a long cylinder about a half mile in diameter. Dammitall, you could take a chunk a half mile in diameter and stretching from one end of the universe to the other, remove it from the universe and let the rest of space curl in to fill up what was missing; and when you were done, no one could measure it. A half mile is a small peanut compared to the immensity of space. Now can I go back to work?"

"In a minute, Steve. What do you know about Maculay's Equations?"

"Maculay's Equations? What do *you* know about them? I mean, what do you *want* to know about them?"

"I'm no abstract mathematician, Steve, but I'm forced to fumble in the dark with some very cockeyed theories that make no sense. Maculay has the idea

that the generation of some sort of negative space would permit gross matter to exceed the velocity of light, but that this negative space would destroy by mutual cancellation this present, or positive space. Does that make sense to you?"

"Y'know what I think?"

"No."

"I think that old saw about the shoemaker sticking to his last is applicable. Stay with your neurones and your pills, witch-doctor, and leave the juggling of space to people who can sight nothing, falling from a vacuum into a void, and explain it."

"Fine," rasped Hanson. "Now that I've been properly roasted for meddling, what gives you to think that no one but an astronomer can think?"

"Steve—if I started to outline medicine to you, it would sound no better than your outline of Maculay's Theories did to me."

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Hanson chuckled. "So we're both stupid, according to the other. Now admitting that I'm stupid and get my income tax fouled up, cannot understand the degrees of infinity, and am completely baffled by the predominance of the value Pi in electricity, do I have a layman's grasp of Maculay's Equation?"

"Barely."

"Then suppose I postulate. Suppose that streak of energy had been a spacecraft passing by at a speed faster than light. And as it passed, its own field of negative space cancelled out a wake of real space as it went."

"That's a fine idea," said Rober. "You might as well postulate that as anything else. Furthermore, the cancellation energy derived might be used to drive the ship; and as far as the loss is concerned, a half mile of space is like bailing Lake Michigan with a teaspoon. The expanding universe is expanding much faster than mankind's puny efforts to trim it down at a half mile per trim."

"Why didn't you tell me this before instead of giving me a lot of guff?" roared Hanson.

"Because the shoe is on the other foot," snapped Rober. "This time you need help. And like the rest of us idiots who show our ignorance when we ask medical questions, you show your ignorance of physics by the damfool questions you ask. But I've done some piddling with Maculay's Equations and the guy has something real and something far above my head, too. Why not ask Maculay?"

"He's not available right now."

"Tough. Probably working on the streak itself, huh? Good thing. He'll get it ironed out. But if you can't get Maculay, get his assistant Redmond. Redmond is a young squirt, but he'll talk if he's urged."

"I've met Redmond."

"Um," grunted Rober. "So that's why you're calling me? Say! Redmond didn't scare you, did he?"

"Sure did."

"Don't let him; Maculay will keep him down."

Hanson decided that this was the time to let the story out. "Redmond came here seeking Maculay. Maculay is on Venus having himself a vacation at my orders, and Redmond wanted him back."

"Wanted him back my foot! Redmond—if anything—wanted to be certain that Maculay was out of the way so that he could plunge into the secret files, using the emergency as reason. What are you doing about it?"

Hanson smiled to himself. "I've done it," he said. "I was just confirming some of my fears by calling you. I've just sent Miss Longacre to get him."

"Pray that she hurries," said Rober. "Redmond is the guy made from the same mold as the Sorcerer's Apprentice."

"You mean the kind of student we used to explain the process of making nitroglycerine to carefully because we knew they'd make it anyway, and blow hell out of themselves if they didn't violate the rules correctly?"

"More'n that," said Rober. "I said Sorcerer's Apprentice and I meant just that. Redmond is the kind of dope who would start manufacturing negative space and not be able to stop the process."

"Someone—or something—has done it."

"Yeah. But they—and Maculay alone on earth—knows what they're doing. And Maculay when I last saw him knew enough to leave it alone."

"Well, it's up to Ava Longacre."

"Hope she's successful."

Hanson remembered the girl's new attitude. "She'll get him," he said.

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Doctor Hanson would not have been able to locate Maculay at all. But he had equipped Ava with the same set of ideas, plus the desire to catch up with the physicist wherever he might have gone; because she was thus equipped, Ava went where Maculay would—and had—gone.

Melaxis, Venus, was a mad mixture of culture and frontier. It boiled with the same sort of teeming millions as New York City; it was a modern city, with white granite buildings, subways, and broad streets filled with racing traffic. But along these broad streets went the rough-shod colonists. They were, for the most part, cut of the same cloth as the colonists of Early America. Men who went to Venus to escape whatever particular hell they felt on earth. Men who objected to taxes, laws, responsibilities, oppressions, regimentations, legalities, religions, and the rest. They were a hardy lot, a bit quick on the trigger and quite inclined to stand upon their own personal integrity. They were just, but their justice was hard-boiled. A man was innocent—or he was guilty enough to get the works.

And it was among this churning metropolis that Ava Longacre landed to seek out Maculay.

Her progress from the spaceport to Maculay was not too arduous, since she knew about where to find him. Ava found a lavish hotel, dragged the bar, picked up a likely-looking character who wanted to visit a gambling hell. Enjoying a chance to show off before this interesting female, the character



took her to a mid-town casino where, he told her, "Mac" was likely to be this night.

"Mac?" she asked.

"Mac is a gambler from way back," he told her. "Luckier than hell."

"Let's go," said Ava.

"That's Mac," he said. It was. Cliff Maculay was sitting before a large card table playing Red Dog. Before him he had a large pile of blue chips, and standing at his elbow watching the pile was a dark-eyed Venusian girl, who swayed languorously to the strains of the music coming from the dance floor next door.

"Would you like to make a hundred?" asked Ava.

"Who do you want killed?"

"Pick up that woman from Mac."

"What's the pitch?" he demanded; "a hundred ain't enough to get me killed."

Ava looked him in the eye. "This is the end of your line," she told him. "If you expect any fun tonight, you'll be better off trying for her, because you're out of a girl friend and Maculay is going to be swapping women shortly."

He looked at Ava, compared her against the Venusian girl in a brazen mental listing of their charms, and repeated a statement made earlier: "Luckier than hell, Mac."

Ava went over to the Red Dog table and stood so that her hip brushed Maculay's arm. Cliff looked up in annoyance, but the frown ceased as he saw her. "Hello," he said cheerfully.

It was obvious that he did not know her, and it was equally obvious that the Venusian girl did not care for the competition. "How are we doing?" she asked.

"Fine," he said.

"Yes, we are," said the Venusian girl, emphasizing her use of the 'we'.

"Cliff will do better now," said Ava.

"The lady knows me," chuckled Maculay.

"Every sharpshooter in Melaxis knows you," snapped the Venusian. "But do you know her?"

Ava laughed. Her voice was a pleasant contralto, throaty, suggestive as she said, "No, he doesn't know me. Yet. Which, darling, gives me an advantage, doesn't it?"

"Don't darling me—"

---

Ava's previous escort was a man of experience, possessor of a fresh hundred, and willing to play the game. His was the simple logic of the wolf; far better to have a woman you might be able to get than one who wanted someone else. Furthermore, he knew enough about human nature to toss a few cupfuls of oil on an already interesting fire. "See here," he said to Maculay, "what's the idea of making passes at my girl?"

Maculay laughed uproariously. He pushed his chair back and stood up, alert. "If she were your girl she'd not be asking me how 'we're' doing," he told the man.

Housemen started to move, slowly, towards the scene of imminent battle.

Ava's escort was willing to start a fire, but he was in no way interested in getting his face pushed in to keep it burning. Yet he could not back out without some show of determination. "I suppose she's *your* girl?" he asked superciliously.

The housemen relaxed. Badinage and billingsgate made noise but it ruined no furniture. The contestants were talking; the kind of fight the housemen were prepared to stop was the kind that took the: "Who—Me?" "Yes, You," Whack! formula which left one of the contenders ready to avenge the lump on his jaw, and willing to use the furniture to do the job.

Cliff relaxed against the card table. "Maybe she is."

"Maybe she isn't!"

"Maybe she'd like to be."

"No accounting for taste."

Ava turned upon her escort coldly. "You haven't any taste. How would you recognize it?"

The Venusian girl knew the situation all too well; she had been looking out for herself for a number of years, and this project included making the best of an opportunity. Her hand strayed behind Maculay.

Then the peacemakers saw something that they were entirely unprepared to stop. Ava Longacre took Maculay by one hand and half-hurled him away from the table, unbalancing him across one hip. Cliff staggered forward—to be caught and supported by his possible assailant. But in the meantime Ava had gone to the edge of the table and had taken the Venusian girl by one wrist. She turned, ducked under the arm, and came up behind in a hammer-lock.

Chips from Maculay's stack dribbled out of the tortured fingers of the Venusian. Ava turned with the girl and hurled her forward into the still-unbalanced men.

The Venusian screamed in anger.

Ava's former escort caught her and kept her from falling, and in doing so he let Maculay slip to one knee.

Someone yelled: "Fight!"

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Hell broke loose. A man clipped his neighbor because the other was luckier than he; a Venusian latched onto a handful of chips from one of the tables and had his wrist broken by the owner of the chips who came down on the arm with a heavy fist. Chips flew through the air and rained down, and many, not caught in the fight, dropped to the floor to pick them up. They got into fights with other gleaners, and the melee spread like a crown fire in a piney woods. Critical mass had been reached, and the fission from civilized human beings to outraged primates spread throughout the room.

Cliff found an ornate chair and separated it to get the back-stringer for a club. The other side was clutched by Ava's escort, who plied it well. Ava

came up between them clutching a small, wicked-looking stiletto and waving it viciously. Maculay slapped it out of her hand.

"Don't start *that*!" he snarled, caving in a likely-looking head with the hunk of chair. He up-ended a table and used it as a protective wall, shoving it forward towards the door. He lost his club over another head and tossed the stub into the face of a third. He splattered the nose of a fourth all over his face, and trampled one fighting pair down to the floor. They paid no attention to him; they had their own private grievance.

Someone yelled: "Police!" and then the lights went out. Maculay steered another course from the door, back through the room full of flailing men and women who were trying now to extricate themselves and make the appearance of innocent bystanders.

Ava opened a door, and the light from inside spilled out over one of the finest barroom shambles ever committed in a high-class gambling hell, where he who wore no evening clothes was not permitted.

Then they were inside.

"Damn," chuckled Maculay. "This is the first time I've ever been inside of a powder room."

"Like it?"

He looked at her. They had lost her former escort in the melee. They had lost some composure, too, and also whatever formality might have been expected.

"Not as well as I thought," he told her. "Where's the hell out?"

Ava pointed to a window.

They left via the window as the door opened. They landed in the gangway between the two buildings, raced for the alley, and ran into a burly man in uniform that stood there stolidly.

Maculay clipped him in a rolling block; the policeman had expected practically anything but a football rush. The pair went down, rolling.

The officer fired one shot at them as they headed into a side-gangway and through to the street beyond. Cliff whistled for a convenient taxicab; they piled into it and were off before the alarm sounded from their rear.

They repaired what damage they could in the taxicab, and carried the rest with them boldly through the finest hotel in Melaxis.

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Once in Maculay's suite, Ava opened her handbag and rolled a horde of chips on the table.

Maculay roared with laughter. "Souvenirs," he chortled.

"Can't you cash 'em?"

"M'lady, you are an angel. You turned up just in time to create a diversion. I got out with a whole skin, anyway."

Maculay looked at her curiously. Her eyes were glowing with excitement; her face was flushed, and she bore that slight dishevelment that brings a beautiful woman down from the pedestal of showcase perfection and makes a warm human of her.

She smiled cheerfully. "What do you mean?"

Cliff stepped to the small bar at the end of the room and mixed two very Herculean drinks before answering. Then he said—after Ava had tasted and approved: "They thought I had the cards marked. I didn't; I was playing a formula."

"But aren't formula players usually losers?"

Maculay laughed. "Baby doll," he laughed, "when you've been trained by the best mathematician in the solar system, you remember the sequence of the cards, evolve a formula of probabilities regarding the shuffling, and then play them according to absolute mathematics. In Red Dog, if there's a Heart Six to beat, each and every card played changes the formula as it lands; if you know your mathematics, you can compute your chances about as well as the Interplanetary Life Insurance Company can compute your expectancy."

"But I spoiled your game."

"That game was ruined anyway."

"It was fun," said Ava, taking a fine pull at the drink.

"A nice shindy, m'lady. And far more better than the game they'd have played once they grabbed me."

"But where will we play tomorrow night?"

"Venus is full of places," chuckled Maculay. "Fact is, the evening is young. Wait'll I collect me a fresh shirt; and I'll have to forget the white jacket since it's a mess. But we can see a bit more Venusian Night Life."

"Done!"

Maculay emerged from the dressing room a few moments later. "By the way, m'lady, what's your name?"

"Ava Longacre."

"I'm—"

"I know. Cliff Maculay."

"Such is fame," sighed Cliff. "You know me?"

Ava nodded. "I've met you before," she said. A faint, subdued recollection of her previous meeting with Clifford Maculay stirred her. She recalled, very dimly, the upsurge of emotion, the pounding of her heart, the complete relaxation of defensive mechanism. Something had been started but never finished, before. Now it was all past, gone, and a new day was yet to be born. "Someone gave me a message for you, but I've forgotten it."

"Maybe we can bring it back," chuckled Maculay. He took her by the arm and led her from the room.



Hanson had committed one pardonable error; pardonable because Hanson, for all of his years and his experience, was no worker of miracles, to whom nothing is hidden, and who can be called omniscient.

For all of his experience in wending his way through the hidden recesses of the labyrinth we call the human mind, Hanson did not know everything and would have been the first to admit this honestly. But he did know that the trouble with both Maculay and Ava Longacre laid in the subsurfaces of the conscious mind. Blocks, inhibitions, and fears instilled as a youth had driven Maculay to seek his excellence in mathematics as a goal rather than as the means to the normal goal of a happy, balanced life. In the filing-cabinet of the mind, however; in the subconscious mind of Clifford Maculay was all of the data of the life he should have led, held there subdued by the blocks of the conscious mind. Hanson had opened the doorway by removing these blocks, and he had done a fine job.

In much the same fashion he had removed the blocks and impediments from Ava Longacre's mind.

Both had suffered from too puritanical an upbringing. In the long distance that lies between white saint and black evil, there is a long dimension lying just below center that is the despair of reformers and do-gooders. This region contains many people and many ideals that are *mal in dictu*. Some impractical reformer had decided, for instance, that liquor is to be abhorred; ergo it is against the desires of society for a man to take a drink. Just one. The idea is, of course, to create a race of saints and Little Lord Fauntleroy sweetness—which probably wouldn't last out the century since the desire to poke someone in the nose for stepping on your rights—or your toe—is the same belligerency that has made mankind fight its way up from the swamps to seek the stars.

Below this region of morals or ethics lies the *mal in facto* behaviour. It is bad in fact and practice to murder, steal, and lie.

Hanson had opened the minds of his pair to the enjoyment of the middle region after a short life of the stilted upper bracket. Like the swing of the pendulum, both Ava and Clifford had dropped about as far as they could go without getting into the truly evil region.

But the doctor's error was in not realizing that the human mind, once released of its inhibitions, can make a shrewd calculation. In the case of Ava Longacre, whose mental blocks would have rendered her undesirable to Cliff Maculay; when once released, the woman's mind reversed its tactics. Where the conscious mind had the balanced life distorted into undesirability, now her mind distorted into undesirability—the more responsible way of living—because she was beginning to enjoy excitement.

All of her quiet life she had been suppressing the love of excitement; now released, Ava Longacre's mind refused to consider the task she had been sent to do; once it was finished, she would be returned to the quiet, unexciting life that she no longer wanted.

So instead of employing her woman's wiles to involve Maculay and bring him back to earth where Hanson could get him to go back to work on his negative space, Ava was helping Cliff cut a wide, rosy-hued swathe through the not-too-holy city of Melaxis.

They consumed a bit more alcohol than was necessary; they danced a bit more close together than would have been called proper at a Boston cotillion, and hazarded sums of money on the roll of a pair of dice or the turn of a card just for the thrill of high blood pressure.

---

It was near dawn when Ava lifted her head from Cliff's shoulder in the taxicab and wiped the lipstick from his cheek with a caressing forefinger. Cliff smiled down at her.

"Baby doll," he said, "let's get married or something."

Ava laughed lightly. "We'll get married—or nothing!"

The sun was above the horizon when Maculay carried his bride—now asleep—over the threshold of his hotel suite. It was late afternoon when the Maculays, man and wife, checked out of their hotel to take a honeymoon in the jungle cities of Venus.

And Hanson fumed and fretted because he had no word from Ava, and worried because he knew that Redmond was poring through Maculay's



secret file of computations and beginning to unravel the data that would permit Redmond to create and establish negative space.

On the third day of such worrying, Hanson knew then he had mis-calculated or over-stepped his reasoning. It was at that moment that Hanson did something that he had stoutly insisted that not even a man should do to his wife, or the reverse. Like reading another's mail, one did not paw through desk drawers nor inspect the corners of another's soul to see whether they concealed something. But Hanson went through the desk drawers of his nurse, attempting to learn how he had erred.

He came up with a small package, neatly tied in a very ornamental manner under the plain store-bag. The name on the fancy ribbon was that of a highly gilded women's shoppe where the salesgirls were very beautiful, the silk very sheer, and the prices very high.

Hanson opened the package. It disgorged a petticoat and bra, through either of which the doctor could have read the telephone directory without his glasses. A scant concession to the custom that a woman should wear lingerie—for the sake of the custom but not necessarily for warmth, protection, fire or famine.

It might have been a gift.

It might have been her own.

It made no difference whether Ava had selected this daring set of scanties for herself or for a gift, wedding or Christmas. It displayed her taste, showed her subconscious desires.

"Damn!" exploded Hanson. "I've been working with a courtesan concealed behind an armor of white starch. Oh, brother!"

The doctor knew. Like two small streams, turned here and there by the minor hills and rocks of fate, they had been joined by Hanson into a flowing river, complete unto itself—themselves—which would go its way as it damn well pleased and overflow its banks to the ruination of anything in its path if it were constrained.

They would not be back until Maculay came back in one year—at which point Ava would subtly change, too, to conform with Maculay's desire.

This left Hanson helpless for one year, during which time Redmond would be working towards destruction with no barriers to his course. Hanson could express no more than an unfounded opinion of the fear of danger; he had neither prestige nor formal education in the field of high-gearred physics. The first objection he voiced would be taken with a nod by whomever official heard it, accepted for what it was: an opinion by a medical savant of seventy years regarding a problem in spacial physics. Then this opinion would be referred to Redmond for official regard. Hanson knew the answer without asking the question. Redmond would laugh in scorn. Redmond would—

Hanson shook his head unhappily. Redmond would be a tough nut to crack. Belligerent, automatically biased against the doctor, any attempt at hypnosis would be fought against most vigorously. Yet—

Jay Hanson had been in his business for a long time, but he had had no challenge such as this for years. And though old in body, he was young enough in mind to contemplate the mental challenge with a certain amount of interest.

He bought tickets and flew to the laboratory site where Maculay and his gang worked on spatial physics; he used his medical prestige as key to admittance, and found Redmond sitting at Maculay's desk checking a huge blueprint of a spacecraft.

---

Redmond looked up. It was obvious that this little scene was one prepared by Redmond. Men who have visitors announced by secretaries, after having signed passes to let the visitor into the inner sanctum, after learning as he must have learned that the famous doctor had come to the huge laboratory site, should not look up from their desks in surprise.

Hanson understood; Redmond was morally right and ethically wrong. He had every moral right to take over Maculay's position during Maculay's absence; that was his appointed job. But ethically, he had no right to paw through Maculay's desk, and take from Maculay's secret files the information that Maculay had forbidden him to see. Now he was play-acting the part of a busy man who had all of the power he needed.

Redmond said: "Yes, Doctor Hanson?"

Hanson paid no attention to the blueprint. "I thought you'd like to know," he said softly, "that I've been unable to locate Maculay."

"Damn!" objected Redmond. Only one who understood what was in Redmond's ambitious mind would know that the disappointment was very false.

"So I came to tell you and also to be curious."

"Curiosity killed a cat," said Redmond.

The doctor laughed. "It's created more kittens than the cats it's killed. Is this still super-top secret or can you let an old man in on it?"

Redmond glowed inwardly at the chance to show off before the doctor. "According to the latest calculations," he said, "the generation of negative space by the force-fields of diagravitic force takes the form of a sphere. Obviously the proper shape for a spacecraft employing one of these generators would be spherical. But we are using a converted spacecraft of the torpedo shape, and I feel that—well, to generate a sphere large enough to enclose the spacecraft in one gulp would produce far too much power. So we are using two of them placed so that their spherical fields produce a pattern something like two equal-sized soap bubbles stuck together. The ship lies longwise through the centers of the circles, since the generators are in the ship, of course."

Hanson nodded. His head bobbed gently, in a measured motion. He was sitting with his back to the room, the window in front of him. He knew that the reflection of the window was in his glasses and that Redmond was watching this spot of light instead of watching the doctor's eyes. Redmond continued to watch as he spoke.

"Within a week we shall have it finished," said Redmond. "Then the stars shall be ours!"

Hanson continued to nod.

"Of course we have not tested the generators as yet. There is no known way of dissipating the energy they develop. Since the realized energy in this real space is sufficient to propel matter faster than the velocity of light, the

outpouring of energy must be paradoxically many times the value of infinity."

Hanson continued to nod.

"This statement, of course, makes no sense," said Redmond, "because of one of the definitions of infinity—which is that number which is larger than the number of all numbers. Here we treat infinity as a definite instead of an abstract, and by our equations we are permitted to multiply infinity by integer numbers and come up with a real answer—in a sort of abstract sense," said Redmond with a slight laugh. It was 'Our Equations' now instead of 'Maculay's Equations'.

---

Hanson continued to nod and Redmond kept watching the spot on the Doctor's glasses.

"However," said Redmond, "the fact is that the power output does not exceed infinity at any time in this space. Not really, and therefore the paradox is answered. It is merely apparent, if you follow me. Actually, the spacecraft is not in real space and therefore it need not have an infinite amount of energy to reach the speed of light. However, there is no way of anchoring the generator on the planet while testing, nor of dissipating the energy. So the only way to test the set-up is to build a spacecraft and take off. If it does not work, we have the standard drivers to get us back."

Hanson kept on nodding. His neck was getting a bit stiff, but he could not stop.

"I've heard the argument that the generators may set up a self-propagating field," said Redmond. "This is so much bosh. The theory that the streak of energy that went through the solar system some weeks ago was the wake of a supervelocity spacecraft seems to be universal among the people who have studied the Equations. Ergo it stands to reason that no destruction of the universe will obtain. We are safe."

Hanson continued to nod.

Redmond smiled quietly.

The doctor said, "You've been working quite hard on this; you must be tired."

Redmond laughed sarcastically. "You've been working harder, Doc. If you've been expecting me to fall under your hypnotic spell with that head-bobbing business, you're getting a stiff neck and no results. You're an old fool with an unfounded horror of anything new. You should view this sensibly; if another race can employ the spacedrive without ruining the universe, so can we. Now why not let busy men alone to work, while you go back to your mental cripples? Good day!"

Hanson fumed but it did no good. He was licked by animosity, disdain, and complete lack of sympathy. There was nothing to do but leave. And the doctor left, half-convinced that Redmond was right in assuming that if one galactic race could use the negative-space drive, another could do the same without fear. But he was only half-convinced; he wanted an opinion from Maculay. There was more here than met the eye.

Some other race knew the secret and were using it. The human race knew the secret and were about to try it. But the man who knew the real answers had gone into a tizzy because of some errata, or factor that was absolutely incompatible with life, liberty, and/or pursuit of happiness.

Hanson grunted. All too often in the case of violent disagreement, all parties were absolutely correct in their own mind, their own honest belief. Maybe this was similar.

One theoretical man feared the results from an abstract analysis of the computations. One mechanically-minded man could not appreciate the possible dangers, but was happy to follow the plans since completion meant fame and fortune for him. Both might be right. But....

---

Hanson shrugged unhappily; it was a bad spot to be in. Yet in the course of his seventy years many problems had seemed insoluble until some factor entered that changed the whole picture. And life itself must have seen many crises, in which the motion of a hand in the wrong direction would have caused the utter downfall of Humanity—or, he thought bitterly, *perhaps we*

*are the result of an ill-moved hand of fate and might truly be great in mind as in work if some prehistoric egomaniac hadn't kicked some unknown prop out from beneath us.*

Perhaps, too, his mind told him, it could have been some half-baked do-gooder trying to help. As he, Jay Hanson, had attempted to help Maculay. The fault was as much his as it was Redmond's. More—Redmond could not help being what he was. Yet, neither could Hanson stand by and see a man go to pieces.

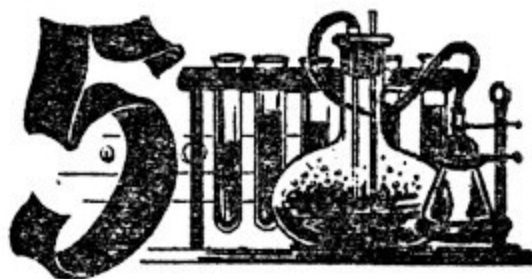
In any case it was not a proposition of fixing the blame; to hell with the blame and the responsibility. Fix it. Fix it. Fix it and forget the fumbling finger that fouled it.

Hanson swore. He was helpless.

Yet for all of his efforts, he believed that something would happen to avert this disaster. It hardly seemed possible that one man's act could destroy the universe. Man's total effort was so puny. Inconsequential. The ignorant savage could not destroy civilization.

But in the back of his mind, Hanson knew that a couple of lumps of plutonium in the hands of an ignorant savage could destroy life beyond the scope of the savage's experience; and mankind's scope was reaching to the stars.

Still fretting, and still hoping for the answer, he headed home.



He was sitting in his office when the telephone rang on the following morning. Hanson answered it slowly, prepared to stall any patient off until he could regain some of his composure and his self-confidence.

"Hanson? Doc, this is Larimore."

"Larimore? Hi. What's up?"

"Doc, this job ain't good."

"What job?"

"*The Black Slash*."

"The what?"

Larimore chuckled. "If that yarn had turned up in the slush-pile, it would have been bounced with a rejection slip. It's not good, Doc. You've got no reason to write that bad, even though you've not written me anything for a couple of years; you don't forget how. But this job sounds like the half-baked efforts of a man convinced that he could write but who lacks the basic fundamentals of story construction. Now—"

"What in the devil are you talking about?" demanded Hanson.

"Didn't you send me a yarn called *The Black Slash*?"

"I—" Hanson paused. Cautiously, he said: "By Edward Lomax?"

"Naturally. That's your pen name. It—"

"Wasn't the job timely?"

"Doc, you ought to know by now that every time something new and frightening comes up, my desk is bombarded with a million stories about it. The best get taken up. That streak of energy a couple of weeks ago has brought fourteen stories so far, and some of them were damned good. But yours—Say, Doc, how come you went to Venus? I thought that you weren't allowed space-flight?"

Hanson paused and shook his head. Edward Lomax was his pen name. It was the pen name supplied to Maculay in the explanation as to why Cliff was in disfavor in the eyes of his fictitious uncle. And it was sort of natural, too, that Maculay would try to write about this thing. But Maculay, either as the renowned Clifford Maculay, or young Cliff Maculay the black sheep, had never written a single line of fiction. Maculay's pedantic papers were full of equations, qualifications, cumbersome sentences, and inverted

phrases—complete with the everlasting 'However' enclosed between commas.

Hanson laughed shortly.

People do not expect a man to step up to his first piano, sit himself down, and run through a faultless repertoire from Bach to Bebop. But these same people nod their heads at a new author's writing and think it is the first time he ever sat down to a typewriter—and then swear that they will do likewise as soon as they get a couple of free hours. Maculay was no exception, plus the fact that Hanson had given his mind the false experience of writing to cover up many irregularities in Maculay's past. Maculay believed he could write and had been writing; actually he knew nothing of the techniques involved. It takes more than a burning desire to see your words in print; it takes at the very least some judgment as to which of your words you select for print plus the ability to produce them in logical sequence. Maculay had tried.

But above all, Maculay had offered a lead—provided unwittingly by Hanson himself. The doctor glowed inwardly, happily. He would now—

"You still there, Doc?"

"You bet. Where did that story come from, Larry?"

Larimore paused a moment. "A small town in the midlands of Venus a couple of hundred miles from Melaxis." Then he exploded. "Hey. Weren't you there? Why didn't you bring it back with you? What the hell goes on —?"

Hanson said, "Larimore, this is a long story and probably a better one than Maculay wrote. But it's important."

"I'm listening. Take off."

The doctor outlined the entire business over the telephone.

"My God," said Larimore. "Now what?"

"Now? It's easy. Send Maculay a special radiogram, addressed to Lomax, stating that Modern Pictures wants the script for a full-length moving picture at some fabulous price providing they can hire the author to rewrite the thing into novel length. You have an option check for five thousand



dollars which will expire within ten days if the author is not present in person at your office before that time."

"I get it. 'Twill be done."

Hanson sat back, relieved; this was the answer he was hoping for. It had come and now all he had to do was to husband his strength until Maculay could get home. Because when Maculay arrived, there would be a big job to do.

---

He spent his time working slowly, resting often. He went to Larimore's office and fitted it with his equipment, on the off-chance that Maculay might be hard to handle. Hanson did not think Maculay would be difficult to re-convert since the true personality was submerged by the false character by mere hypnotic suggestion. It should be remarkably easy. But the doctor wanted to take no chances.

He read Maculay's sorry attempt at fiction. It was not good fiction but it interested Hanson because there was so much fact concealed in its descriptive passages. Maculay, unable to think too deeply about the negative space concept, or real and unreal space, and variable-matrix wave mechanics, had treated the whole scientific formulation with a touch of the ridiculous. Just as Cliff, upon hearing of the streak of energy, had laughingly included it in a 'story' because he was hypnotically unfitted to treat his opinion as anything but fiction-fantasy, he was again concealing the truth behind a thin disguise. It was all there.

All there, Hanson saw with a sour finality, but the solution. Maculay had pulled the old gag of having the fabulous machine totally destroyed, complete with its secret. A poor gag, and unfitted for modern writing, especially unfitted for application to fact. For, in fact, this was not a story; it was the truth, told by a man who must tell it as fiction since the truth literally hurt him. But there was no true solution, and once the negative-gravitic generators were started, the unreal root of negative space would spread to engulf the universe.

This 'story' of Maculay's convinced—or rather pinned the last doubt down—Hanson that his guess-work was right. But handing such a story to any official as true data would get the doctor nothing but a horselaugh—at the least—and possibly a trip to the looney-bin for observation.

However, he would have the truth at hand soon enough. Maculay would know what steps to take.

Even if Maculay ordered everything to stop, while the answer was found.

---

Hanson was working in Larimore's office when Maculay came in with his bride.

The doctor looked at them both; he nodded affably.

"Doc!" roared Maculay cheerfully. "What in hell are you doing here?"

"Came to kiss the bride," said Hanson. "And she looks lovely enough to kiss."

"Go ahead," said Maculay; "I'll permit you eight seconds."

Hanson smiled at Ava, but shook his head. "I've got one more thing on my mind," he said quietly. "Cliff, what do you know of Maculay's Equations?"

"He's an uncle of mine," started Maculay. "He came up against a tough one. He found a way to exceed the speed of light—but doing it would destroy universal space by a sustained and spreading cancellation. It—"

"Maculay, what would you do if you were *The Clifford Maculay*?"

"Go fishing."

Hanson touched a button at his elbow. There was a soundless flash of brilliant light as the photoflash bulb planted in the desk lamp flared. Then as Maculay stood, tense with shock, Hanson said, in a forceful tone: "Clifford Maculay, the hypnotic suggestion that I gave you before must cease. I order it to stop; I order you released. You are once more Doctor Clifford Maculay, who must—"

The jovial smile faded from Maculay's face. The twinkle in his eye changed to a calculating glitter, and the lines of Maculay's face hardened. "Hanson," he snapped, "what has been going on?"

"You've been on vacation," said the doctor. "And while you were traipsing all over Venus, Redmond has opened your secret file and is starting to build a supervelocity spacecraft. You must put a stop to it."

Maculay looked startled for a moment. Then he said: "Redmond is a pompous sort of juvenile jackass, I admit, but he isn't that stupid."

"I've seen his installation."

Maculay shrugged. "I'm not a jealous man, Doc. I've had my day; I've done my work; I've laid my cornerstone. I've even been stumped. Now if Redmond can solve the problem that had me licked, I'll be the last man on earth to deny him his triumph."

"Clifford, from all I've heard about this, total destruction will result if any man energizes a volume of negative space."

"Quite right," said Maculay. And as he said it, his eyes clouded and he winced gently.

"Redmond has added nothing to your calculations."

Maculay stood up with a dry smile. "As a physician you are Number One on earth. As a psychiatrist you are tops. I know what you've done and it's been good. I hope," he added slyly, "that she likes me as well this way as she did the other way—or can you change her too? Or," he continued with growing comprehension, "is it 'change her back, too?'"

"Back."

"Good. So as a witch-doctor you're tops in any jungle. But as a physicist, you don't know a gravitino from a vocal fricative."

"Meaning?"

"Meaning that before any judgment is cast, I shall have to see the evidence."

Hanson stood up. "So it's back to the laboratory site."

Maculay nodded, held out an arm which Ava took happily, and then he said: "And from the lab site to the stars, Doc."

Hanson grumbled: "Or total extinction."

Maculay did not hear him. He was looking down at Ava. "Doc," he said slowly, "you'd better come along. Snooky, here, needs to be slowed down to my level and you're the guy to do it."

Hanson did not tell Maculay that Ava's reconversion would take no longer than his own; the doctor wanted to be in at the end of this, good or bad. He merely nodded, then waited while Maculay made arrangements to fly to the laboratory site. His name worked wonders; an official plane was being warmed up by the time they left Larimore's office and headed towards the airport.

---

Redmond greeted them with a hearty smile. Only Hanson, who had every reason to doubt Redmond's happiness at Maculay's return, saw the falsity of the greeting. Redmond, of course, was on a spot; yet, the man was convinced of his own correct reasoning, and this justified his acts. Redmond's greeting was less hearty to Hanson; obviously Redmond would have preferred to deal with Maculay alone. Having the doctor there might be awkward, for Maculay might be talked into belief, whereas Hanson was more than likely to ignore the words and their import, and deal entirely upon whether the sayer of the words was lying or telling the truth.

Redmond believed in a swift attack.

Once the original greeting was over, he plunged in: "We spent some time trying to locate you as soon as it became evident that the energetic streak that went through the solar system produced the sort of radiation that we had been theorizing over," said Redmond. "Lacking Maculay, I was asked to open your secret file and see what could be made of it."

"You discovered the trouble, then?"

"Yes."

Cliff relaxed. He had been under a strain visible only to Hanson; the doctor nodded. When a man is in a mental tizzy because he's hit upon an insoluble dilemma, it makes no difference who solves it. The weight of strain went out of Maculay; the mental run-around that had kept him fighting to the exclusion of everything else was gone. The couple of months of rest had done wonders; now the final true release from strain added to it. Give Maculay another few months of absolute freedom from strain, and Cliff would be ready to take on the world with a hand tied behind him.

But Hanson knew there was trouble ahead, for, unless he were very incorrect, Redmond was bulling it through and—

"You've discovered the error?"

Redmond laughed. "Your equations showed that negative space cancelled real space."

"Yes. And I could figure no other way."

"This is true in limited cases," said Redmond. "The consensus of opinion is that the streak of energy was nothing more than the mutual destruction of a cylinder of space being cancelled by the passage of a spacecraft enveloped in a spherical field of negative space. Upon working with that theory in mind and applying other bits of true evidence gained from the readings and measurements of the streak, we have solved your dilemma."

"Let's see our equations," suggested Maculay.

"Rather," said Redmond, "let's visit the spacecraft."

"All right."

Hanson said: "Are you certain that you're not assuming too much?"

"Meaning?" asked Redmond coldly.

"You are basing everything on the fact that an alien spacecraft passed through here. How do you know?"

Redmond laughed in a superior manner. "Since matter cannot exceed the velocity of light without being encased in a volume of unreal space—and since a volume of unreal space would kick up the same sort of wake as we measured—we can assume that some intelligence has developed negative

space and is using it. Negative space, Doctor Hanson, is not to be found free in nature."

"But you've really added nothing to Maculay's Equations."

"We've proven by observation that the sustained destruction of the universe will not obtain; we'll prove it, too."

Hanson snorted. "This isn't a game of bridge," he said; "you're not bidding a grand slam just to see if you can make it without the ace of trump."

"But we know that it has been done. Nothing more need hold us up; we know!"

Hanson added another page to his mental notes regarding Redmond. Frustrated genius, second rater really, Redmond was the type of man who had always been protected against danger. In the course of his life, he had never faced the consequences of one of his own acts; therefore he fully believed that every time he was about to step off of the deep end, some Divine Providence would save him. If Redmond were permitted to do as he wanted to do, it was "Sign to Redmond" that he was on the right track. Some people call it superstition; some call it intuition; some call it foolishness. To Redmond, it was a sort of Fate.

---

Maculay stood up and led the way to the doorway. "Let's look at this," he said.

"Cliff," said Hanson, "nothing has changed since you went away. Real and unreal space are still mutually destructive. And if you couldn't figure it out, no other man on earth could."

Redmond said, "True, at that time. But we've had extra evidence to work on."

"But—"

"Forget it," said Redmond; "we know what we're talking about."

Maculay entered the control room of the ship first. He looked it over with interest, then nodded. "Everything is in ship-shape fashion," he said.

"We could start tomorrow if we had to."

Maculay looked at the controls that projected side by side on the polished black panel labelled *Upper* and *Lower*.

"Dunno," said Maculay thickly.

Hanson watched him carefully. "Cliff," he said quietly, "you knew about the streak of energy, too. If that were the answer, you'd have come out of your mental tizzy."

Maculay turned to Redmond. "What means have you to prevent the sustained reaction?"

Redmond shook his head. "We don't need any. If another race can do it—"

"Don't be an idiot! Just because one race makes iron steamships it is no sign that iron floats on water."

"But it stands to reason—"

"You'd bank your life on it?"

"Yes."

Cliff Maculay took the two handles, one in each hand. His eyes glazed a bit, and he laughed uncertainly. "Maybe the creation of the universe was started by some fool who created negative space," he said thickly. "You simple idiot, this is exactly the danger that almost drove me nuts; you haven't solved a thing!"

Maculay stood there, watching Redmond. Then the frown left his face, and his body tightened. His eyes lost the hard glitter and took on a luminous air, which became half-humorous and half cynical as the corners of his mouth quirked up.

Hanson took a deep breath. Maculay the physicist had become Cliff Maculay the hell-raiser in just that short a time, because he was once more faced with the insoluble.

"No!" yelled Hanson.

But Maculay laughed. "Might as well wreck it," he jeered. "Better to wreck this fool's work than destroy the universe. Damn idiot, Clifford Maculay. Better—"

Maculay slammed the *Upper* switch to the right.

"—let the ne'er-do-well foul it right!"

Maculay slammed the *Lower* switch to the left.

There was a perceptible shift in the frame of reference, a hiatus in the solidity of things, like the rug slipping on the polished floor, like the fancy movable steps in the Fun House, like the bottom of the quicksand lake, like space itself being warped.

Then Arcturus passed the nearby viewport in a single flash of blue-white, and seconds afterwards a second star flashed, then a third.

"The backward sort," chortled Maculay. He was neither Maculay the Physicist now, nor was he Maculay the hell-raiser; he was a glad mixture of both. "It came to me," he told Hanson, "just like *That!*"

"But—?"

"Easy. Easy. The output from the upper generator creates negative space. But before it can establish an expanding field, the output of the lower generator nullifies it. For the lower generator is making a field of positive space equal to the force from above. Take unity. Add unity and you have two. Cancel one unity and you have unity—plus a spacedriver!"

---

Thirty thousand light years away, a recorder wiggled, a bell rang, and a sentient creature came out of a quiet complacency with a roar. Then came the clangor of a huge alarm and other creatures came tumbling into the huge room. They watched the recorder anxiously; then as it levelled off they left, slowly. Six remained; the others got into a small spacecraft and took off. There would be no nova for the suppression squad to extinguish; all that was needed was for the safety squad to go there and teach newcomers how not to play with fire.



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